Lakshay Tyagi

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Software Developer and Research Engineer building cutting-edge AI solutions at AWS
My work revolves around building Search, Large Language Model (LLM), Computer Vision, Machine Learning, Retrieval-Augmented Generation, and Distributed Database Solutions

Education

New York University, Courant Institute of Mathematical Sciences — Sep 2022 - May 2024
MS in Computer Science; GPA: 4.0/4.0; Fellowship for exceptional academic performance — New York City, NY

Indian Institute of Technology, Kanpur (IITK) — Jul 2017 - May 2022
BTech, Double Major Electrical Engineering; GPA: 9.5/10.0; General Proficiency Medal — Kanpur, UP

Work Experience

Amazon Web Services (AWS) - Software Development Engineer - AI/ML — Jul 2024 - PRESENT

Intelligent Table Analysis Agent

- Collaborated with Amazon teams to gather requirements and design workflows for table reasoning and code-execution agent
- Partnered with scientists to translate prompts and agent behavior strategies into production-ready implementation via Strands SDK
- Developed a secure, isolated code-execution service using AWS Lambda to execute agent-generated code and return responses dynamically
- Integrated the execution framework into agent orchestration pipelines, improving overall model accuracy on tabular data by 25%

Bring Your Own Model (BYOM) Integration for Bedrock Knowledge Bases

- Designed and implemented support for customer-provided and SageMaker-hosted Machine Learning models within Bedrock Knowledge Bases
- Built scalable model hosting and invocation workflows using SageMaker endpoints and Rust-based Text Embeddings Inference (TEI) containers
- Developed load testing and endpoint scaling strategies to validate performance, optimize throughput, and ensure reliable large-scale inference
- Collaborated with internal stakeholders and customer teams to define onboarding, invocation, and model compatibility requirements, improving overall retrieval accuracy by 20%

Agentic Retrieval and Metadata Aggregation

- Implemented agentic retrieval workflows with inferred metadata aggregations, enabling dynamic operations such as sum, average, and grouped analytical computations over metadata
- Developed intelligent retrieval pipelines that inferred aggregation intent from user queries and orchestrated metadata-aware execution flows
- Enhanced retrieval relevance and reasoning quality by integrating aggregation-aware context construction into LLM response generation
- Improved overall LLM response accuracy by 10% through metadata-driven retrieval optimization and agentic query execution logic

Hierarchical Chunking For Managed Knowledge Base

- Improved retrieval and storage for hierarchical chunking, storing text in a **NoSQL database**, and reducing latency by 200 ms
- Optimized ElasticSearch clusters for enterprise search with a reduction in latency by 47% and savings of >150,000$ per year

NYU Langone Institute - Research Engineer - Publication Oct 2023 - May 2024

- Developed Large Language Model (LLM) solutions for labelling medical reports with pathologies with explanations
- Pretrained BERT-based models on a corpus of 3,400,000 medical reports with Masked Language Modelling
- Engineered Coreset-based Active Learning algorithms to select a corpus of 11,000 reports for labelling with a teacher model
- Trained LLAMA-2 using LORA with distributed GPU training in **PyTorch** to improve Macro F1 score to 0.89

Internships

Bayer, AI Team - Data Science Intern Jun 2023 - Sep 2023

- Orchestrated time series forecasting pipelines in Databricks for predicting sales growth, integrated with SnowFlake using SQL
- Engineered features and trained XGBoost, LSTM (RNN), and graph models to reduce MAPE to 0.11 on backtests
- Developed a web application in Flask for visualizing forecasts for >100 million dollar sales for 20 product segments

Mitacs Globalink - Research Intern - Publication Jun 2021- Aug 2021

- Set up a distributed training framework in PyTorch for UNet-based federated learning models for brain tumor segmentation
- Achieved Dice Similarity Coefficient of 0.674 using variable local tuning of client parameters with learning rate scheduling

Samsung Research Institute - Software Engineering Intern May 2020 - Jul 2020

- Implemented Autoencoder-based deep learning models in TensorFlow for denoising low light, grainy videos
- Delivered State of the Art SSIM of 0.89 and PSNR of 30.14 for video generation using perceptual loss and sensor input

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Technical Skills

Programming Languages: Python, Java, C/C++, Ruby, Scala, TypeScript; **Databases:** DynamoDB, ElasticSearch, Snowflake, Postgres
Cloud Technologies: AWS, Azure, DataBricks, Kubernetes, Hadoop, Spark, Grafana **Software:** MATLAB, R
Libraries: PyTorch, Tensorflow, Keras, NumPy, Scikit-Learn, OpenCV, pandas, HuggingFace, Spring/Boot, cvxpy, Scikit-Learn, NLTK

Publications

- Humor@IITK at SemEval-2021 Task 7: Large Language Models for Quantifying Humor and Offensiveness A Gupta*, **L Tyagi***, A Pal*, B Khurana*, A Modi: Publication in Association of Comp. Linguistics (ACL) : ACL-ICJNLP 2021
- Federated Learning Using Variable Local Training for Brain Tumor Segmentation: A. Tuladhar, **L. Tyagi,** R. Souza, ND. Forkert, BrainLes MICCAI 2021